

SASWAT K. NAYAK

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EDUCATION

Master of Science, Machine Learning in 3D Photonics, University of Wisconsin-Madison.

August, 24

RELEVANT COURSES

Software Engineering, Image Processing, Advanced Machine Learning, Statistics, Information Retrieval, Databases, Probability and Stochastic Processes, Regression and Time Series Analysis, Machine Learning in Materials, Algorithms and Data Structures, Linear Algebra, Graph Databases, Econometrics.

SKILLS

Technical Skills	Javascript, Python, CI/CD, Ruby on Rails, Docker, Gitlab, Golang,R, C++, SQL, Tableau, Clinical Trials, Spark, GPU, Typescript, Angular, React, Java, Agile, KPI, GCL, Mage, Azure, Caboodle, A/B testing, Excel, Power BI, GraphQL, Node.js, EHR Data.
Machine Learning	NLP, GANs, RNN, Gen AI, Tensorflow, PyTorch, GANs, Airflow, Statistics, LLM, Pandas, Numpy, MLOps, Spacy, Large Language Models(BERT, GPT-2, T5, RoBERTa), Google TTS, GBM, XGBoost, ETL Pipelines, Flink, Langchain, Transformers, Kusto.

EXPERIENCE

Projects [Live Demo](#), Click to see dashboards!

- Execute an end-to-end data engineering project on real-time stock market data using Kafka, Amazon Web Services (AWS), Apache Kafka, Glue, Athena, and SQL.
- Utilized LangChain and Llama2 to develop a chatbot fine-tuned on my experiece. Enhanced its performance by 70% through the implementation of RAG, embeddings, vector stores, and optimized document chunking techniques.
- Engineered a high-performance file-system backed by relational databases, introducing a user-friendly web interface for comprehensive CRUD operations and authentication.
- Analyzed Lyft data and made GIS dashboards using various tools and technologies, including GCP Storage, Python, Compute Instance, Mage Data Pipeline Tool, Streamlit, BigQuery, and Looker Studio.

Data Scientist(Full Time) Innovaccer Analytics(Digital Tech Unicorn)

July 2020-July 2022

- Convinced the clinical leaders to shift the knowledge base ontology from relational db(PostgreSQL) to graph db(Neo4j). Shifted the entire knowledge base DB(PostgreSQL) containing mappings to ICD, SNOMED codes to a graph database.
- Led a team of interns to extract patient data from progress notes, EHR data, achieving a 90% reduction in time required and a 50% decrease in associated costs by implementing NLP solutions(NLTK, Spacy, RoBERTa) and using auto-encoders for anomaly detection.
- Built statistical analysis dashboards using streamlit for hospital ratings, per capita income, readmission rates, major diseases, and number of insurance claims across different locations
- Orchestrated the automatic deployment and scaling of a containerized task queue subsystem responsible for batch processing across dozen of worker nodes- Docker, EC2.

GRA: Gen AI in 3D Photonics, Physics

September 2022 - Present 2023 — *Madison, WI*

- Introduced novel spatial descriptors for 3D lattices that captured more information than the traditional symmetry based descriptors and outperformed them in machine learning tasks by 50%(F1 Score).
- Used a mixture of generative models capable of generating 3D lattices with a wide band gap.

Machine Learning Intern, IISc

November 2020 - December 2020

- Built bioinformatics pipelines for analysis of complex genomic or phenotypic data for HCV virus dynamics.